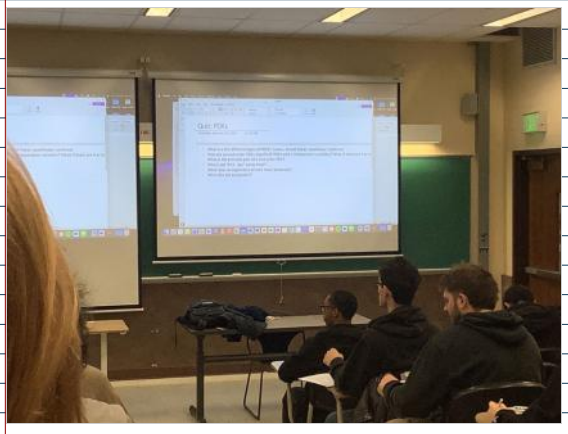
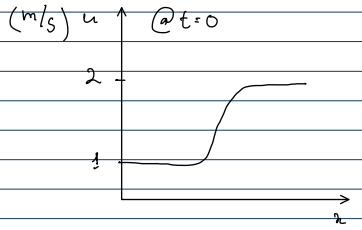


$$\frac{\partial u}{\partial t} + \underbrace{u\frac{du}{dx}}_{\frac{\partial}{\partial x}Q(u)} = 0$$

$$Q = \frac{u^2}{2} \quad ; \quad u_f = \frac{Q(u_2) - Q(u_1)}{u_2 - u_1}$$

Geometry, operating conditions

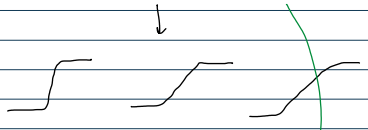


Given this, is this going to have weak solution? (for $t \to \infty$)

↓

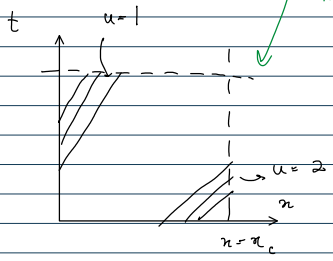
No, because it can't catch up! u is ???

vs.



Solutions through characteristic lines.

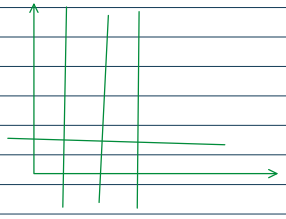
BC's : $u(x=0,t) = 1$



The lines can be drawn directing from this

↓

These are exact solutions.



$$\frac{(u_i^{n+1} - u_i^n)}{\Delta t} + \frac{(u_i^n)^2 - (u_{i-1}^n)^2}{2\Delta x} = 0 \quad \Big\} \text{ explicit}$$

$$\frac{(u_i^{n+1} - u_i^n)}{\Delta t} + \frac{(u_i^{n+1})^2 - (u_{i-1}^{n+1})^2}{2\Delta x} = 0 \quad \Big\} \text{ implicit solutions.}$$

Numerical BC's

$$\frac{\partial^2 u}{\partial x^2} = 0$$

Second order PDE's :

- Elliptic → every point affects everything else (steady state potential flow)
- Hyperbolic → propagating solution up (allows weak solns)
- Parabolic
  - ↓ normal
  - ↳ broad
  - (one zero eigenvalue)

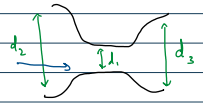
  } must specify BC at every point

Q3 (HW1) → no refinement
(discretize PDE's, BC's, system of non-linear eq's)



1.) Criteria for continuum

$$K_n = \frac{\text{mean free path}}{\text{characteristic length}} \lesssim 0.0001$$



might have to look at various BL's

2) Thermodynamic equilibrium

Reversibility, assume no gradients (uniformity) → things are infinitely slow

Why need thermo? →
→ define fluid properties (what are measurables)

} Can assume local equilibrium

→ Conc. + temp grad } coupling effects (one affecting the other)

3. Mechanical energy